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Data Science

Diagnox – Explainable AI for Personalized Healthcare via multi modal fusion

Group No. 6

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Project Guide
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Abstract

Healthcare systems face challenges in timely and accurate diagnoses due to the complexity of EHRs, imaging, and genomic data, which are often analyzed in isolation. Existing AI models act as "black boxes," reducing trust among clinicians. DiagnoX proposes an explainable, multi-modal AI framework that fuses diverse healthcare data for unified predictions. Using PyTorch/TensorFlow, it applies deep learning (CNN, ResNet, DenseNet, Bi-LSTM) and transformer-based NLP models (BERT, ClinicalBERT) for disease detection and patient profiling. With explainability tools like SHAP and LIME, DiagnoX ensures transparency, personalization, and trust — enhancing early diagnosis and treatment strategies.

Sustainable Development Goals(SDG) mapped

SDG 3: Good Health & Well-Being

- Provides accurate and early diagnosis using multi-modal healthcare data.
- Supports personalized treatment through explainable AI insights.

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Introduction

Growing patient numbers and the rise of chronic diseases are placing increasing pressure on healthcare systems to deliver fast and accurate diagnoses. At the same time, large volumes of medical data such as electronic health records, imaging, lab reports, wearable data, and genetic information are being generated but are often not effectively integrated. As a result, doctors may rely on limited data, which can lead to delayed or less accurate diagnoses. Although artificial intelligence has shown strong potential in analyzing complex medical data, many existing systems lack transparency and explainability, reducing trust among healthcare professionals.

Objectives

1. Develop deep learning models to integrate EHRs, medical imaging, and genomic data for accurate and comprehensive diagnosis.
2. Implement transformer-based NLP models to extract insights from unstructured EHR text for patient profiling and early detection.
3. Apply advanced architectures to analyze medical images and genomic sequences for precision medicine.
4. Incorporate explainable AI to provide transparent, interpretable predictions that build clinician trust and adoption.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	MedPix 2.0: A Comprehensive Multimodal Biomedical Dataset for Advanced AI Applications	Irene Siragusa, Salvatore Contino, Massimo La Ciura, Rosario Alicata, Roberto Pirrone	2025	Built MedPix 2.0 dataset from MedPix®, curated via semi-automatic + manual cleaning, stored in MongoDB; Retrieval-Augmented VLM (DR-Minerva) with Knowledge Graph integration using Llama 3.1.	Dataset still limited to MedPix scope; lacks real hospital-scale data and diversity.
2	Explainable Multimodal Deep Learning in Healthcare: A Survey of Current Approaches	B. Dhanalaxmi, PVVS Srinivas	2025	Survey of multimodal deep learning approaches in healthcare, focusing on feature extraction, fusion, and explainable AI techniques for medical diagnosis.	Survey is high-level; lacks experimental validation and depends on secondary sources.
3	Clinically Accessible Small Multimodal Radiology Model for Report Generation	João M. Z. Chaves et al	2025	Small open-source VLM for chest x-ray report generation in resource-limited settings.	Smaller model lags SOTA on complex findings; mainly tested on CXR.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Masked Modeling for Joint Medical Image-Text Embedding	Z. Chen et al.	2024	Joint masked image modeling + masked language modeling to learn a shared embedding space for retrieval/zero-shot.	Depends on quality of pretraining corpus; noisy text reduces performance.
5	A review of Explainable Artificial Intelligence in healthcare	Zahra Sadeghi, Roohallah Alizadehsani, Mehmet Akif CIFCI, Samina Kausar, Rizwan Rehman, Priyakshi Mahanta, Pranjal Kumar Bora, Ammar Almasri, Rami S. Alkhawaldeh, Sadiq Hussain	2024	A systematic review of XAI techniques (feature-based, surrogate, concept-based, local/global) applied to healthcare.	Lacks standard evaluation, limited clinical validation, and unreliable post-hoc explanations for complex medical data.
6	Multimodal Fusion Models for Pulmonary Embolism Risk Stratification	Noga Cahan et al.	2023	Intermediate fusion of CT features with clinical variables using bilinear attention + TabNet.	Task-specific; requires external validation.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Med-UniC: Unifying Cross-Lingual Medical Vision-Language Pretraining	Zhongwei Wan et al.	2023	Cross-lingual vision-language pretraining on multilingual CXR-report pairs with tri-perspective objectives.	Still centered on CXR; linguistic/domain shift issues remain.
8	CheXzero: Expert-level Detection from Unannotated Chest X-rays via Self-Supervised Learning	Eric Tiu et al.	2022	Self-supervised pretraining on unlabeled CXRs, enabling zero-/few-shot pathology classification.	Focused on CXR; generalization to other modalities/workflows unproven.
9	Artificial Intelligence-Based Methods for Fusion of Electronic Health Records and Imaging Data (Scoping Review)	Farida Mohsen, Hazrat Ali, Nady El Hajj, Zubair Shah	2022	Review of 34 studies on AI fusion of imaging and EHR. Categorized fusion strategies (early, late, joint). Compared ML/DL methods for clinical outcomes (diagnosis, prediction).	Review-based; no new model proposed; highlights lack of public multimodal datasets.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	Integrated Multimodal Artificial Intelligence Framework for Healthcare Applications (HAIM)	Luis R. Soenksen, Yu Ma, Cynthia Zeng, Leonard Boussioux, Kimberly V. Carballo, Liangyuan Na, Holly Wiberg, Michael Li, Ignacio Fuentes, Dimitris Bertsimas	2022	Proposed HAIM framework; trained 14,324 models on HAIM-MIMIC-MM dataset across multiple modalities (tabular, time-series, text, images). Used Shapley values for modality importance.	High computational cost; requires large multimodal datasets not widely available.
11	ExAID: A Multimodal Explanation Framework for Computer-Aided Diagnosis of Skin Lesions	Adriano Lucieri, Muhammad Naseer Bajwa, Stephan Braun, Muhammad Imran Malik, Andreas Dengel, Sheraz Ahmed	2022	Developed ExAID framework using Concept Activation Vectors (CAVs) + Concept Localization Maps (CLMs) to generate textual + visual multimodal explanations for skin lesion CAD.	Focused only on melanoma vs nevi; may not generalize to broader dermatology cases.
12	Fusion of Medical Imaging and Electronic Health Records with Attention and Multi-Head Mechanisms	Cheng Jiang, Yihao Chen, Jianbo Chang, Ming Feng, Renzhi Wang, Jianhua Yao	2022	Proposed multimodal attention module + multi-head GMU to fuse imaging and EHR data using CNN (3D ResNet). Evaluated on Glasgow Outcome Scale and Alzheimer's datasets.	Performance depends heavily on dataset size; limited evaluation on few diseases.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition	Shih-Cheng Huang, Liyue Shen, Matthew P. Lungren, Serena Yeung	2021	Contrastive image–text pretraining aligning local regions with report words; uses attention for global+local alignment.	Needs paired reports; mostly chest-x-ray data; limited generalization.
14	JoImTeRNet: Joint Learning of Chest X-ray and Radiology Reports	Chufeng Van Uden et al.	2021	Joint image–sentence and region–word matching with attention; combines cross-entropy and triplet losses.	Requires paired data; mainly evaluated on CXR.
15	ConVIRT: Contrastive Learning of Medical Visual Representations from Paired Images and Text	Yuhao Zhang, Hang Jiang, Yasuhide Miura, Christopher D. Manning, Curtis P. Langlotz	2020	Bidirectional image–text contrastive pretraining on radiology images + reports to learn transferable visual features.	Requires large-scale paired text; strongest on CXR; less tested in other modalities.

Research Gap(Limitations of existing systems)

1. Single-Modality Limitation: Most diagnostic systems use only one data type, reducing accuracy and missing the full clinical context.
2. Lack of Explainability: AI models act as black boxes, making it hard for doctors to interpret and trust results.
3. Data Quality Issues: Incomplete and inconsistent healthcare data lower model reliability and accuracy.
4. Prediction Bias: Region-specific or limited datasets cause biased and unfair diagnoses.
5. Low User Adoption: Lack of trust, transparency, and user-friendly design limits real-world use.

Problem Definition

- Current healthcare AI systems face fragmented data sources and poor clinical workflow integration.
- Lack of explainability leads to black-box predictions, reducing trust among doctors and patients.
- Existing solutions offer limited personalization, often giving generic recommendations.
- Privacy and regulatory compliance remain major challenges.
- There is a need for a multi-modal, explainable AI system that fuses EHRs, imaging, and genomics for accurate, transparent, and personalized insights.

DiagnoX aims to bridge this gap by delivering trustworthy, interpretable, and patient-centric healthcare solutions.

Scope

1. Advancement in Explainable AI: Further development of more sophisticated XAI frameworks, tailored specifically for medical use cases, will help provide deeper interpretability of model decisions. This can strengthen clinical trust and meet regulatory standards.
2. Integration with Cloud and Edge Computing: Leveraging cloud-based AI services and edge computing devices can make the system more accessible and efficient, enabling AI-powered healthcare even in resource-constrained or remote environments.
3. Precision Medicine and Personalized Healthcare: By combining genomics, imaging, and EHR insights, the project can evolve into a precision medicine platform that provides individualized treatment recommendations, identifies potential drug responses, and supports preventive healthcare initiatives.

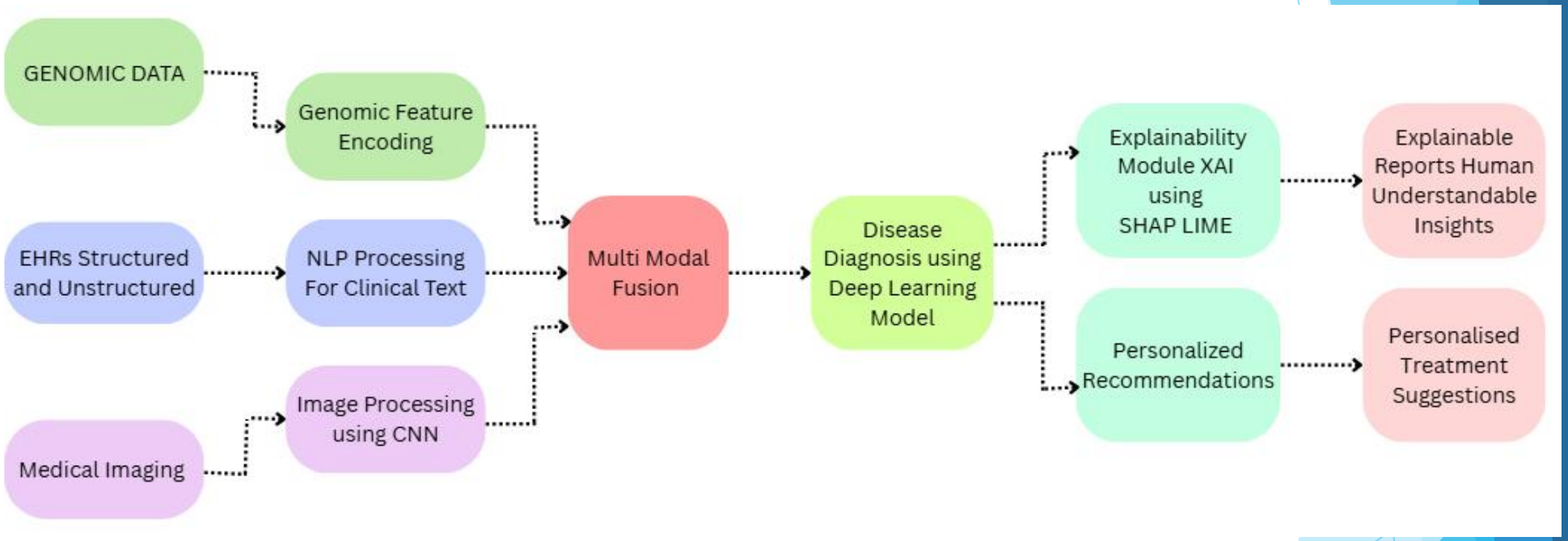
Technological Stack

- **Frontend:**

- HTML5: Structures the doctor-patient portal, chatbot forms, and report interface (Current version: HTML5, W3C standard).
- CSS3: Provides responsive, mobile-first layouts, accessibility, and multilingual font support (Current version: CSS3).
- JavaScript (ES2023): Enables interactivity, chatbot UI, and medical data visualization updates (Current version: ECMAScript 2023).

- **Backend:**
- Python 3.12: Core backend engine powering AI/ML models and API services (Current version: Python 3.12, Oct 2023).
- Custom CNN (PyTorch-based)
- NLP (Symptom Analysis): Pretrained biomedical NLP models like BioBERT and ClinicalBERT.
- scikit-learn (1.5): Supports preprocessing, classical ML baselines, and evaluation metrics.
- Grad-CAM (for generating heatmaps). SHAP - LIME, Captum: Explainable AI libraries for interpretable model predictions.

Proposed system architecture/Working/Module Diagram



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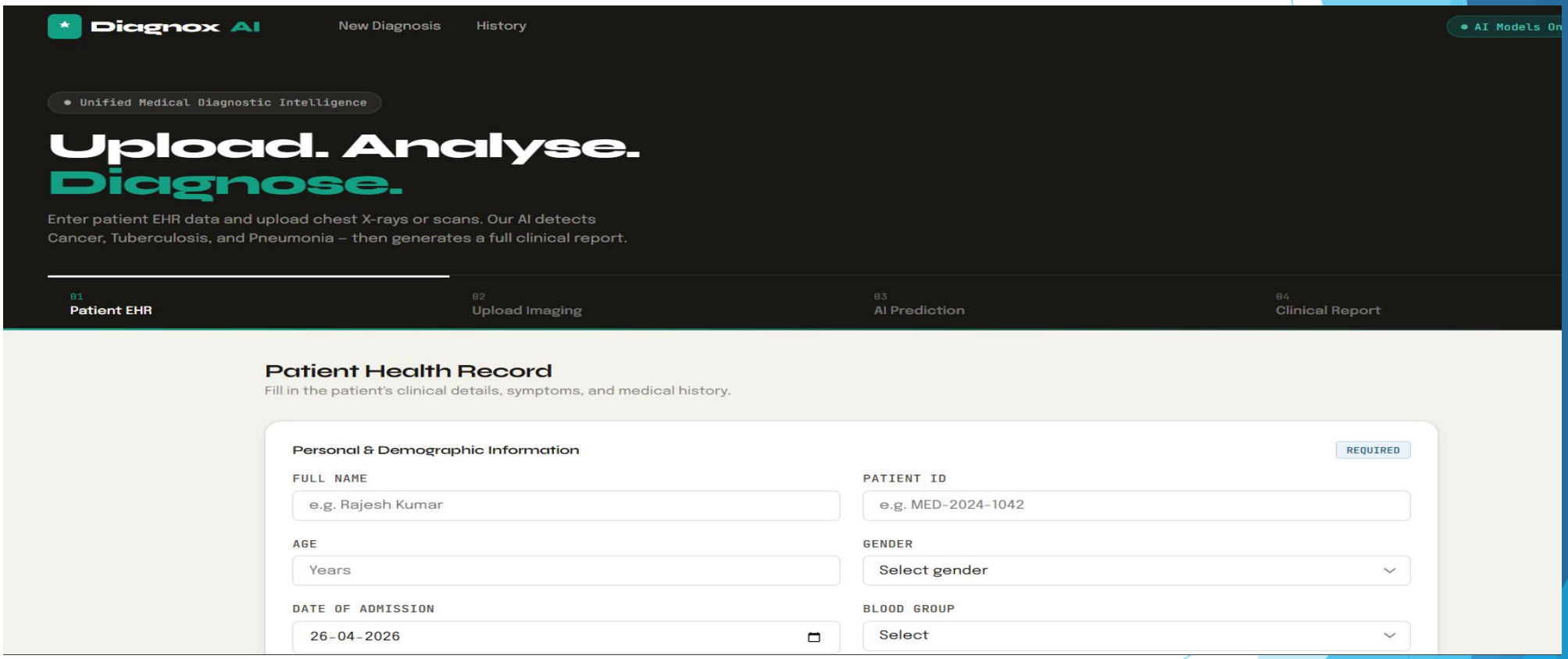

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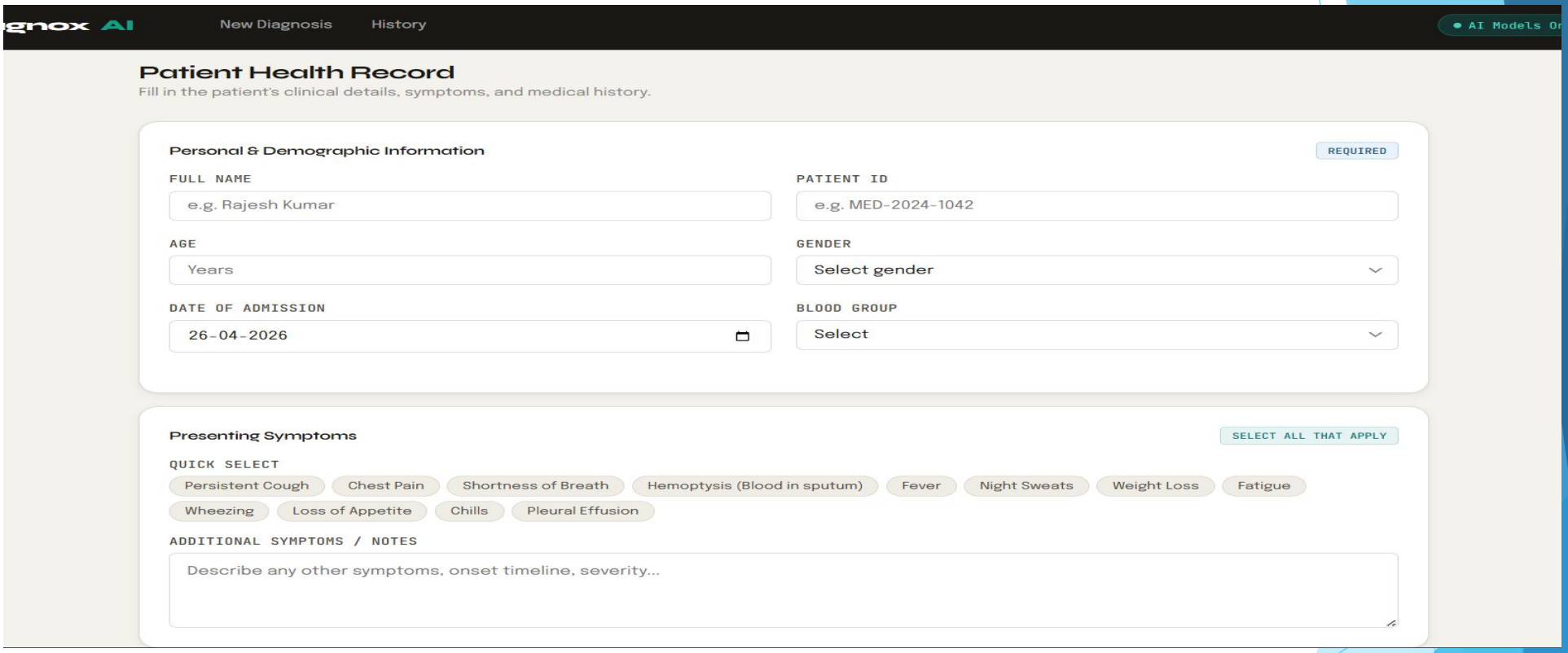

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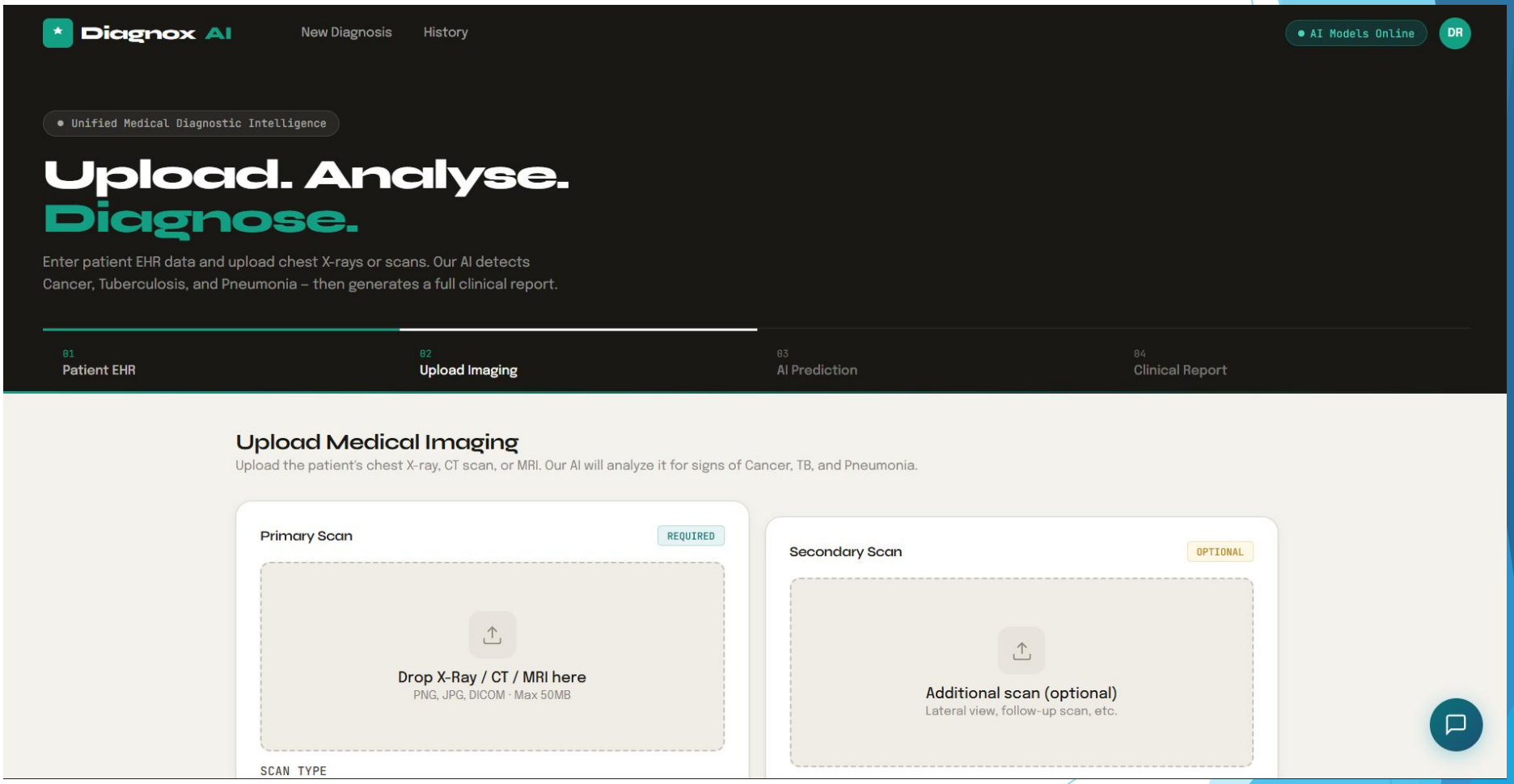

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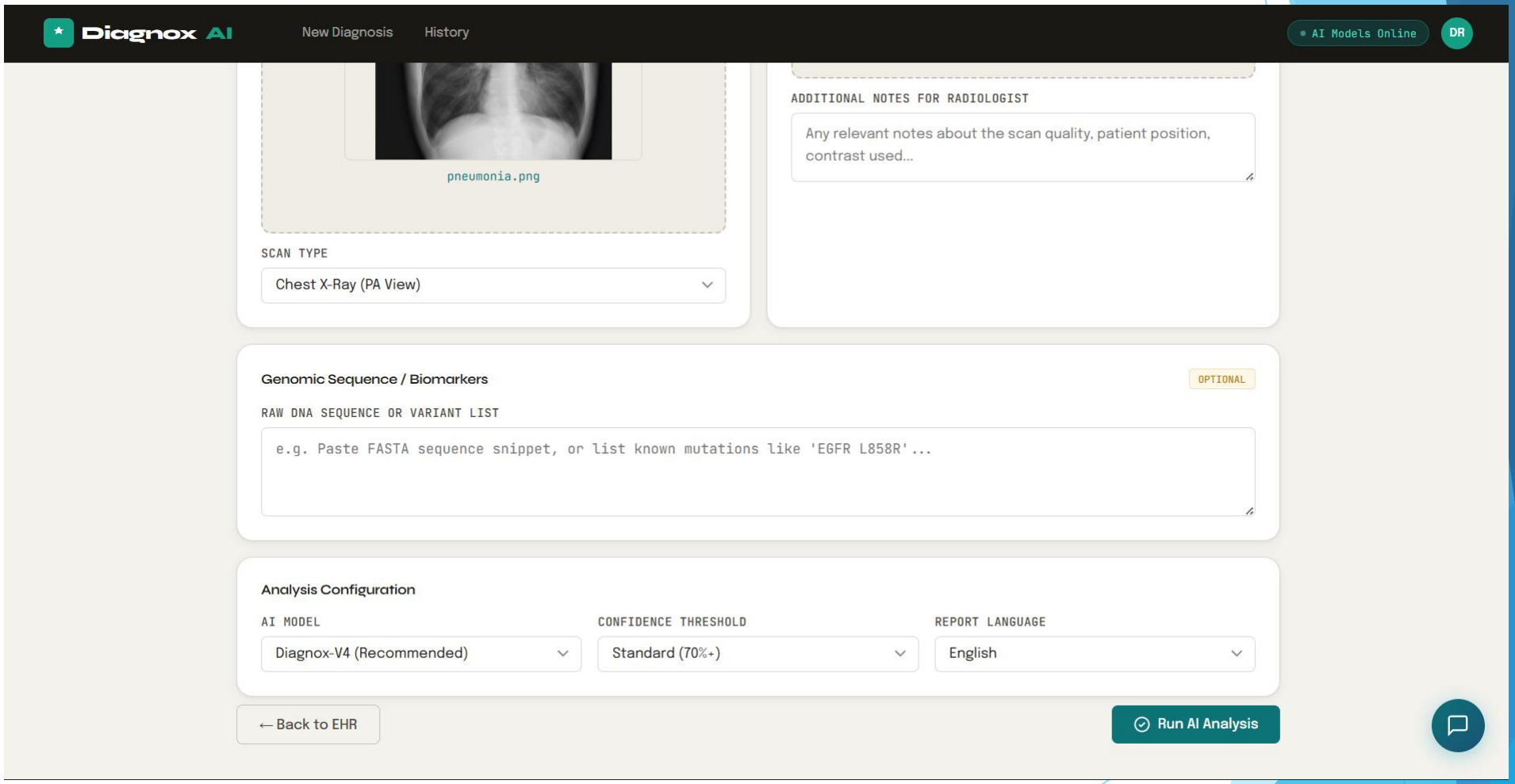

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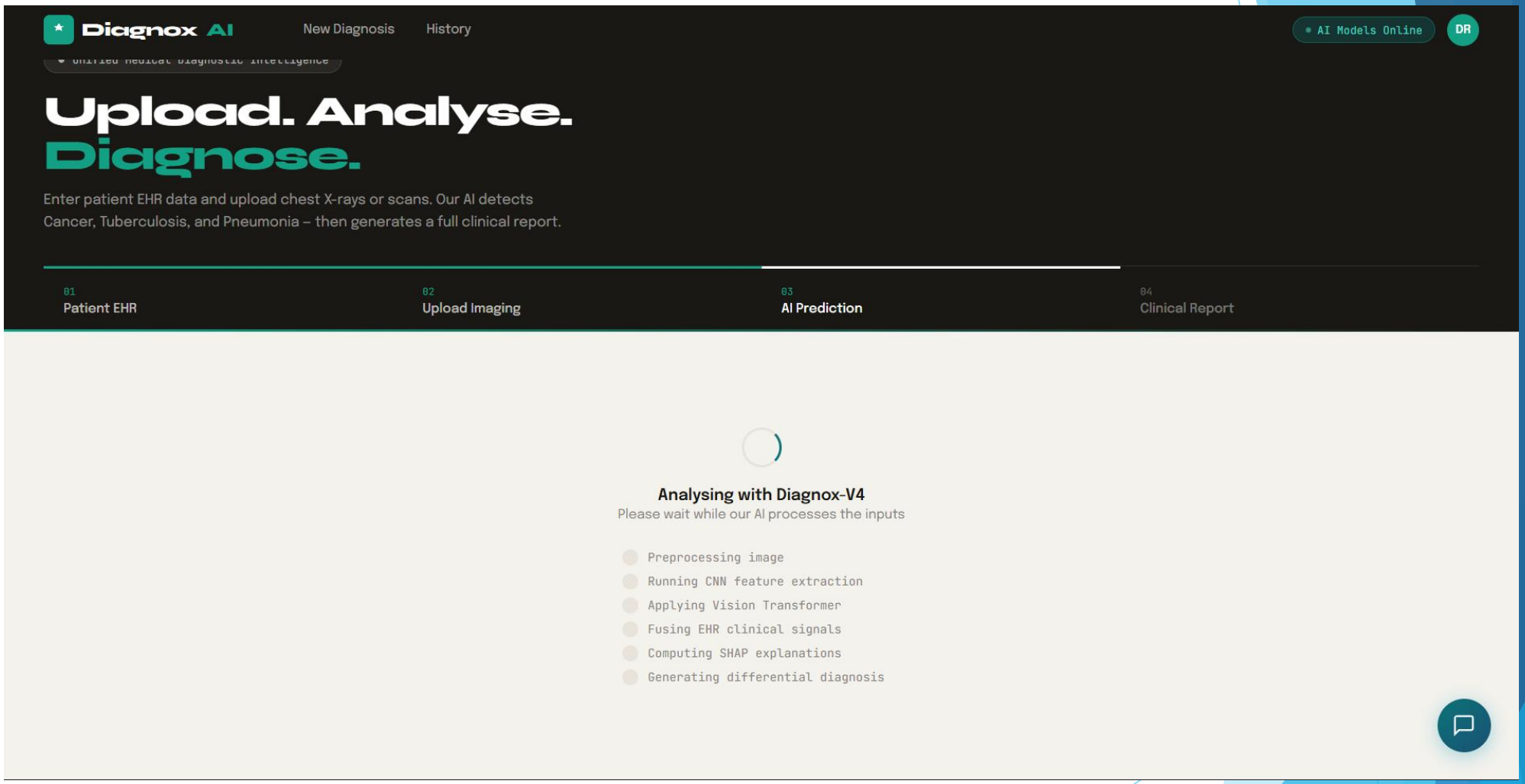

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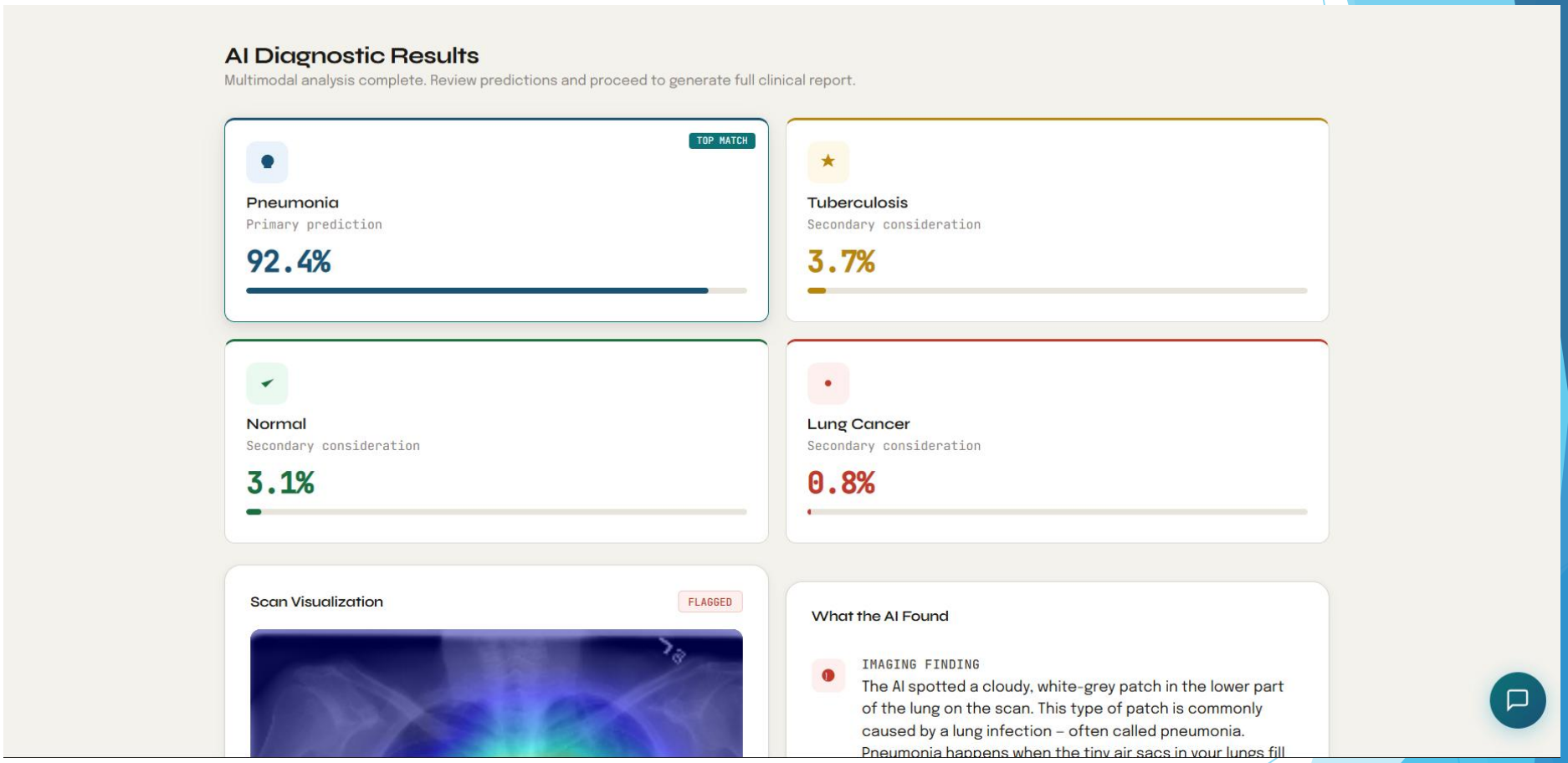

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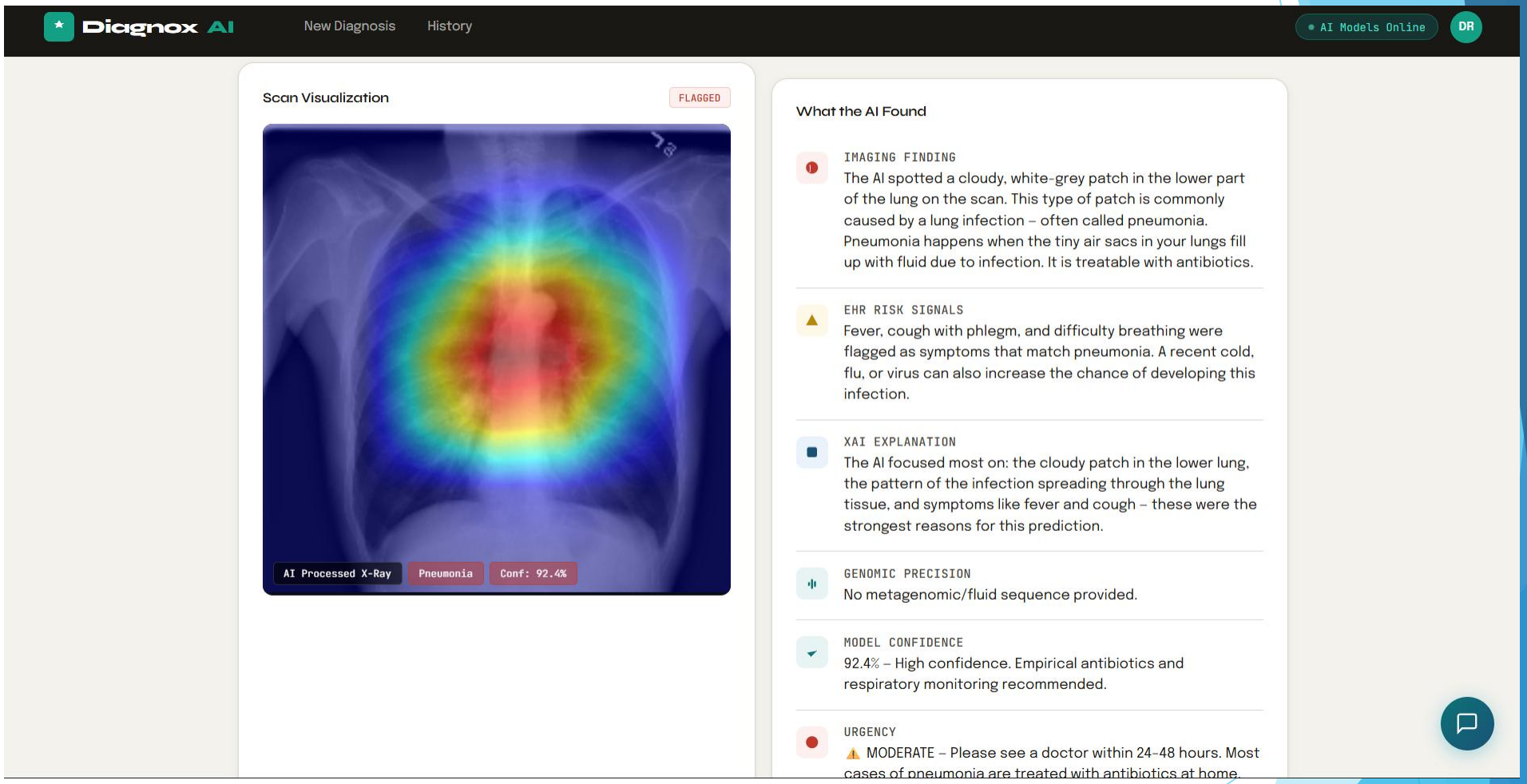

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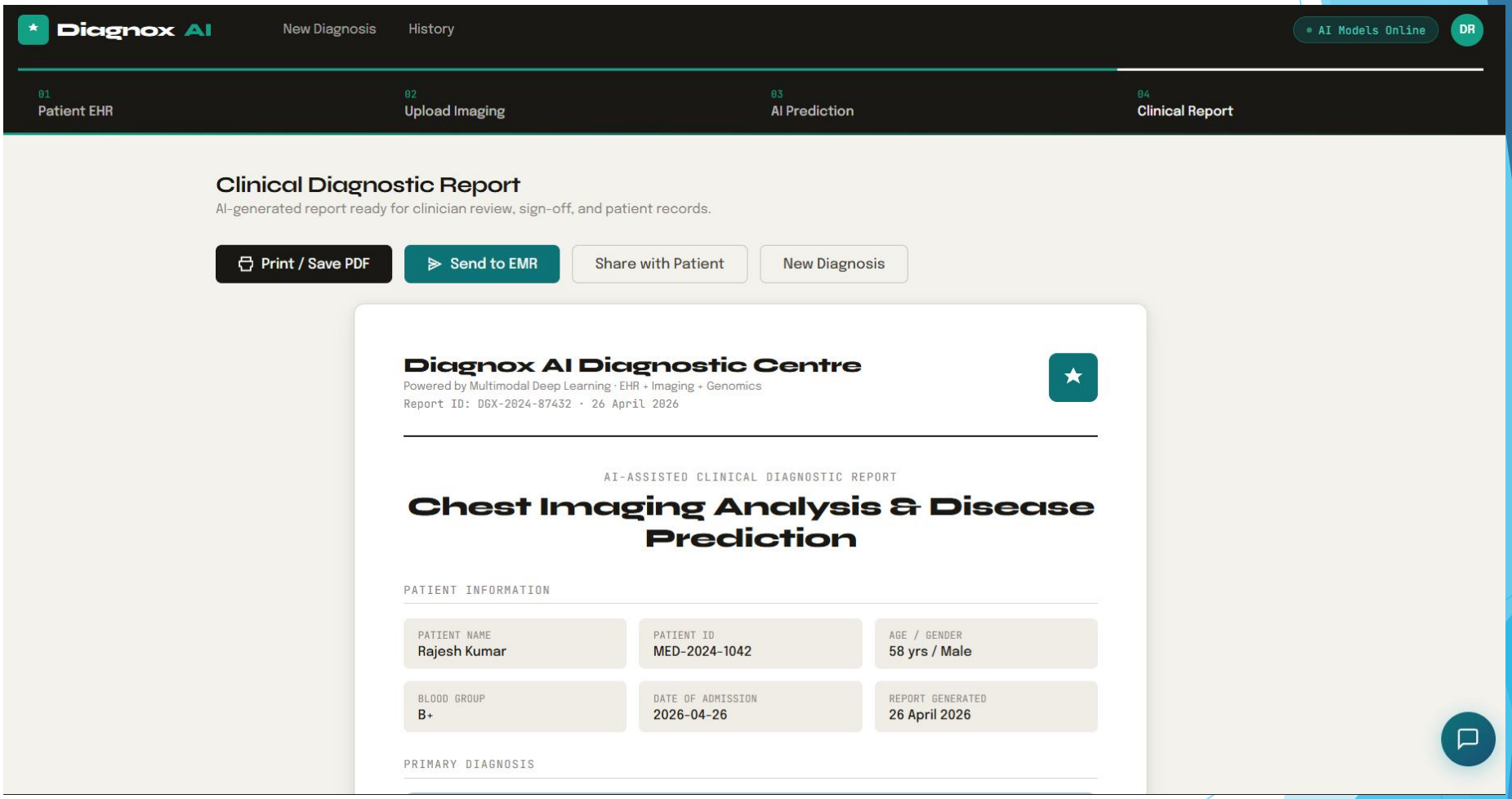

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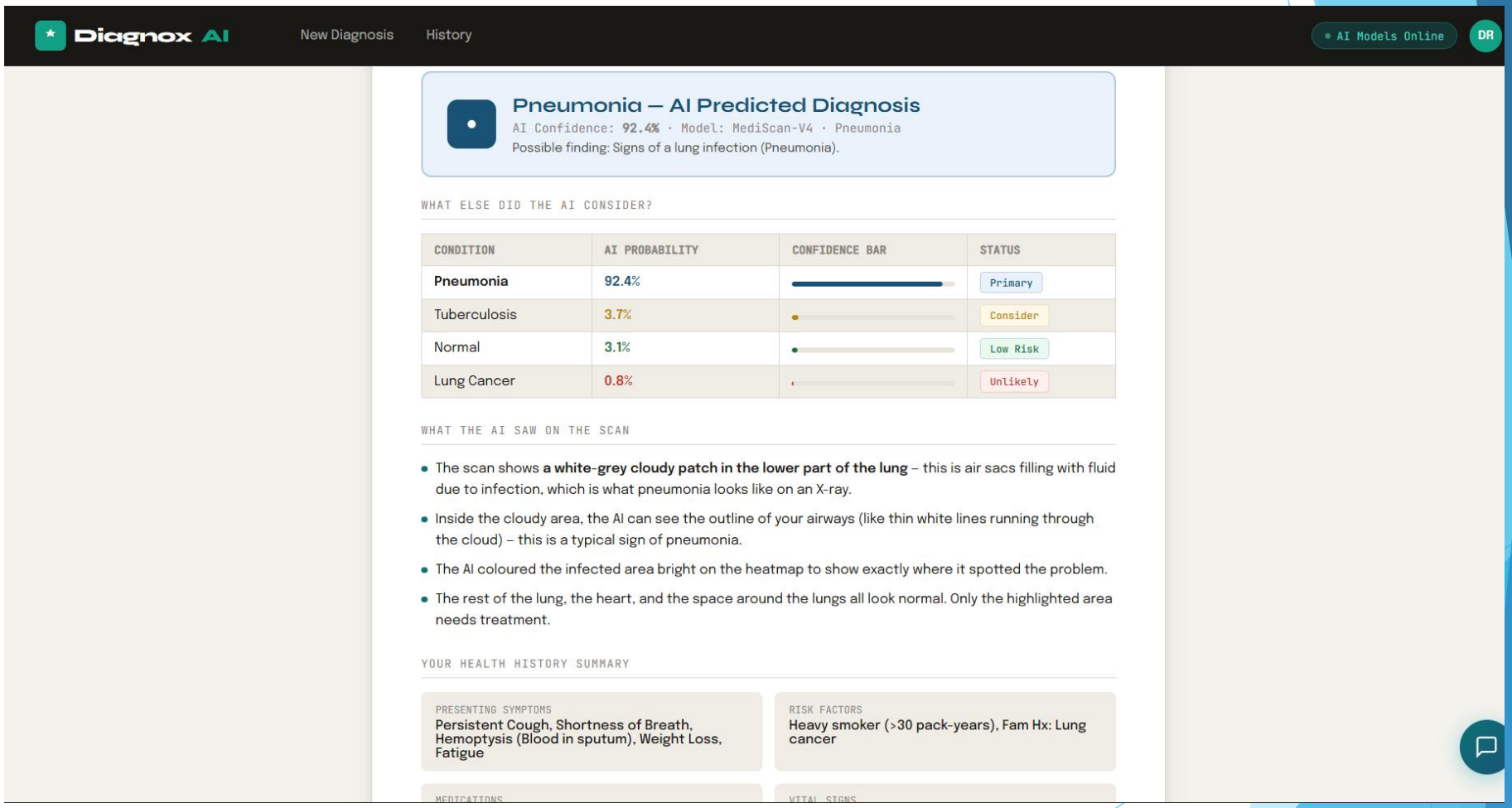

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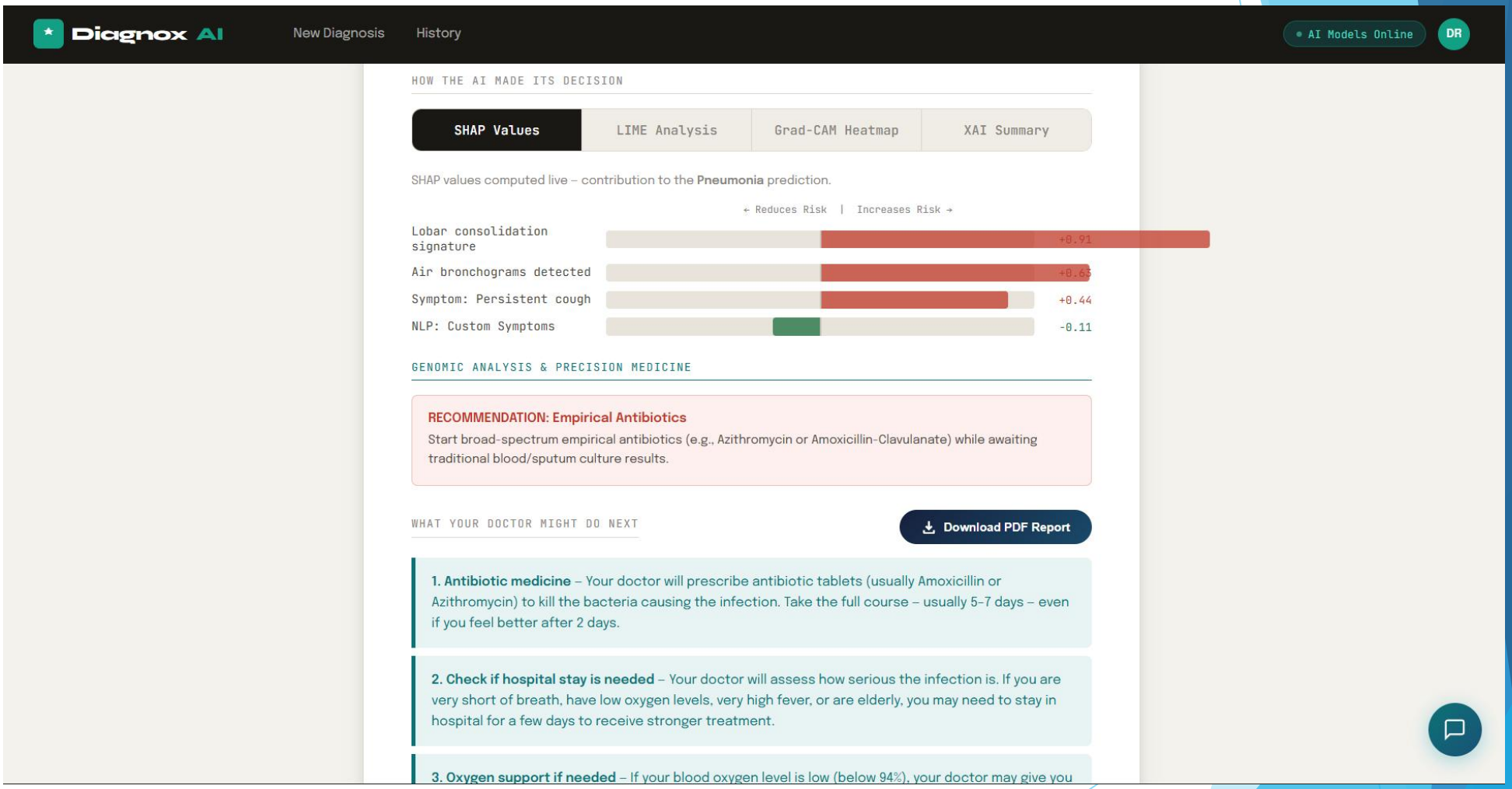

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Thank You...!!